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An Explainable and Lightweight CNN Framework for Robust Potato Leaf Disease Classification Using Grad-CAM Visualization

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ABSTRACT

For identifying foliar diseases in crops at an early stage, accurate detection is necessary in maintaining food security, minimizing economic losses, and cultivating sustainable agriculture. In staple crops, potato is highly vulnerable to lethal diseases like Early Blight and Late Blight that can drastically affect both the quality and the quantity of the yield. Conventional diagnostic procedures using visual observation and/or laboratory examinations are frequently tedious, time-consuming, and susceptible to error. To address these problems, in this research, we propose a novel deep learning architecture using a customized convolutional neural network (CNN) for classifying potato leaf images into three distinct classes, namely Early Blight, Late Blight and Healthy. The model is trained on a selective and heavily augmented subset of the PlantVillage dataset containing 11,593 images and further optimized using regularization techniques like dropout and batch normalization. The system architecture is intended to keep the tradeoff between performance and computational efficiency, so as to fit real-world agricultural scenarios. To increase interpretability and improve trust, we use the Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize the regions in space of the leaves that most contribute to the prediction of the model. The experimental results show superior performance and the proposed model reaches 99.14% accuracy and close-to-perfect precision, recall and F1-scores in all of the classes. Grad-CAM visualizations validate that the model is robust in attending to biologically meaningful regions for the disease symptoms. In addition, we perform comparative analyses against recent state-of-the-art models, and demonstrate that the proposed approach outperforms the others in accuracy and interpretability.

1 | Introduction

Agricultural productivity forms the backbone of food security and economic stability in many countries, especially in agrarian economies like Bangladesh [1]. Among various staple crops, potatoes (*Solanum tuberosum*) hold a prominent position due to their adaptability, nutritional value, and global demand [2]. However, the yield and quality of potato crops are significantly

hampered by various foliar diseases, most notably Late Blight (Figure 1a) and Early Blight (Figure 1b), which can devastate entire fields if not identified and mitigated in a timely manner [3]. These diseases, caused by the pathogens *Alternaria solani* and *Phytophthora infestans* respectively, present visually similar symptoms in their early stages, such as necrotic spots, chlorosis, and irregular patches, making them difficult to differentiate even by experienced agronomists [3, 4].

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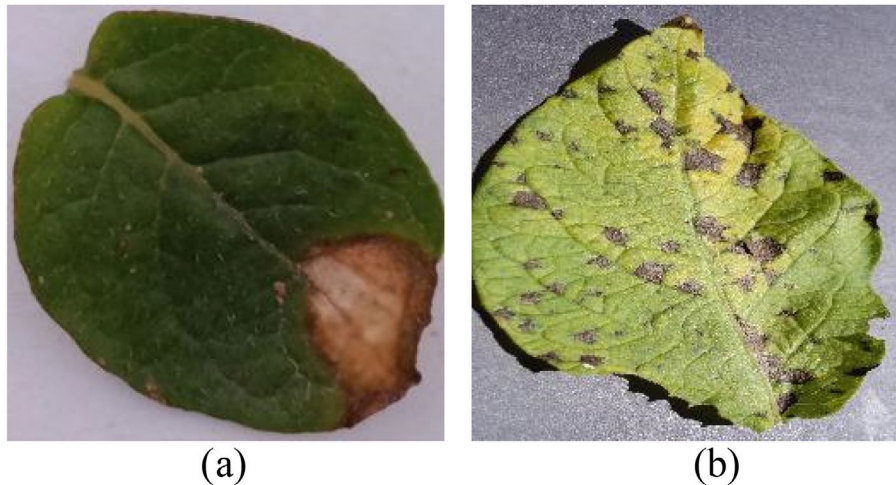


FIGURE 1 | Visual examples of potato leaf conditions, illustrating typical symptoms of (a) Late Blight and (b) Early Blight.

Conventional techniques for disease diagnosis, which include visual inspection or laboratory analysis, suffer from the underpinning subjectivity, the limitation of time, and employing experts in the process [5]. Moreover, they are impractical for large-scale or real-time monitoring in field conditions, especially in resource-constrained regions [6]. Therefore, there is a trend to move away from manual detection and classification of crop diseases in leaf images towards using AI and computer vision, in particular deep learning, to automate the process [7, 8].

Among the deep learning approaches, CNNs have emerged as the state-of-the-art for image classification tasks due to their capability to hierarchically learn and extract spatial features from raw image data [9–11]. Several CNN-based studies have shown high classification accuracies in plant disease detection. However, most existing models either rely on overly complex architectures such as VGGNet, Inception, and ResNet, which are computationally expensive or lack transparency in their predictions, rendering them less useful in real-world agricultural decision-making, where explainability is essential [12–14].

In addition to accuracy, model interpretability is a crucial requirement for building trust among end-users such as farmers and agronomists [15]. A model that merely outputs a classification label without explaining the rationale behind its prediction is of limited value, especially in high-stakes domains like agriculture [16]. This has led to a growing interest in Explainable AI (XAI) techniques, among which Grad-CAM has gained prominence. Grad-CAM provides intuitive visual explanations by highlighting the regions in the input image that the model considers most influential for its prediction, thereby enhancing transparency and trust [17].

There remain, however, a few challenges. First, most public datasets for plant disease detection are narrow in focus, which cannot fully reflect the real-world changes in the light, background, disease status, etc., such as the datasets limited to one kind of crop, and the datasets that do not take sufficient real-world variability into account, including the light condition, disease progression, background variations [18, 19]. Second,

due to imbalanced data and small numbers of image samples, the overfitting that happened is obvious and generalization results are poor [20]. Finally, there is a lack of addressing the important requirement of light weight as well as interpretable models for real-time agricultural applications on mobile or edge devices [21].

This paper fills this gap by introducing a customized CNN-based deep learning approach to differentiate three diseases of potato leaf, namely, Early Blight, Late Blight, and Healthy. The model is trained on a curated and augmented subset of the PlantVillage dataset to ensure diversity and representativeness. The proposed architecture is computationally efficient, avoiding unnecessary complexity while being designed to balance accuracy with computational efficiency. Furthermore, to improve interpretability, Grad-CAM is added to the pipeline, so as to provide visual explanations of how the model makes decisions.

The key contributions of this research are as follows:

1. We propose a custom-designed CNN architecture tailored for potato leaf disease detection, optimized for both performance and computational efficiency.
2. We introduce a comprehensive preprocessing and data augmentation strategy to enhance model robustness and improve generalization across diverse conditions.
3. We integrate Grad-CAM into the framework to provide post hoc visual explanations of predictions, thereby improving transparency and fostering user trust.
4. We conduct a comparative analysis with recent state-of-the-art models, demonstrating that the proposed approach achieves superior accuracy, reliability, and explainability.

The remainder of this article is organized as follows: Section 2 provides a review of related work in plant disease classification using deep learning; Section 3 outlines the proposed methodology in detail; Section 4 presents the experimental results, evaluation metrics, and analysis; Section 5 concludes the study and outlines future research directions.

2 | Literature Review

The integration of deep learning into agricultural diagnostics, particularly for plant disease identification through image-based classification, has garnered significant attention over the past decade [22]. Among the various crops, potato (due to its global economic and nutritional importance) has become a focal point for researchers aiming to automate disease recognition using computer vision techniques [23]. This section reviews the recent literature on deep learning-based plant disease detection, with a specific emphasis on potato leaf classification, CNN-based model architectures, dataset limitations, and explainability in artificial intelligence. The section also outlines the prevailing research gaps that motivate the development of the proposed framework.

2.1 | Deep Learning in Plant Disease Detection

CNNs have revolutionized plant pathology by enabling automated feature extraction from raw images, eliminating the need for handcrafted features. Mohanty et al. [24] were among the first to demonstrate the viability of deep learning in this domain, using AlexNet and GoogLeNet on the PlantVillage dataset to classify 26 diseases across 14 crop species with high accuracy. However, their approach was primarily based on transfer learning from large, generic models, which are often over-parameterized for domain-specific applications like potato leaf disease detection.

Subsequent studies focused on tailoring CNNs for specific crops. For instance, Amara et al. [25] utilized a shallow CNN for banana leaf classification and achieved promising results, though their model lacked generalization capability across varying environmental conditions. More recently, Shaheed et al. [26] employed ensemble learning by combining ResNet-50, VGG-16, and InceptionV3 for multi-class crop disease classification, achieving accuracy upwards of 96%. Nonetheless, such ensemble architectures are computationally intensive, making them unsuitable for deployment on edge devices or in resource-limited settings.

In the context of potato leaf disease, Singh et al. [27] explored the fusion of CNN with pretrained VGG networks, attaining a classification accuracy of 97%. Similarly, Bharti et al. [28] combined ResNet with CNN to classify leaf diseases and reported 98.36% accuracy. Although these models perform well in controlled environments, they often lack transparency, leaving users without an understanding of the model's decision-making logic.

2.2 | Dataset Constraints and Generalization Challenges

The majority of existing plant disease classification models are trained on datasets such as PlantVillage, where the input images are clean laboratory-captured with a consistent background and controlled lighting. Although such datasets are an important starting point for model development, they do not encapsulate

the challenges of true agricultural settings which may include varying lighting conditions, diverse leaf angles, background “noise,” as well as the different stages of the disease [29, 30]. As described by Sharma et al. [31] and Sowmiya et al. [32], models trained only on controlled datasets tend to overfit and may not generalize well to natural settings. Even though research with Kalaivani et al. [33] highlights the increasing relevance of augmented and synthetic data generation in overcoming this gap, however, there still currently exists no widely accepted methodology for creating labeled training data that reflects the variability found in real-world agricultural contexts.

2.3 | Explainability in Plant Disease Classification

Despite the growing body of research focusing on accuracy, relatively few studies incorporate Explainable Artificial Intelligence (XAI) mechanisms into their frameworks. This omission is critical, as stakeholders in agriculture, including farmers, agronomists, and policymakers, require interpretable outputs to make informed decisions [34, 35]. Grad-CAM has emerged as a popular post hoc interpretability tool for CNNs. While it has been applied in fields like medical imaging and autonomous driving, its integration into agricultural diagnostics remains sparse [36]. Fahim et al. [37] initiated the application of Grad-CAM in plant phenotyping, yet its use in potato leaf disease detection, particularly in conjunction with custom lightweight CNN models, has not been extensively explored. The absence of visual justifications for predictions reduces the trustworthiness of AI-based systems in agriculture. As Abdelhamid et al. [38] argue, incorporating visual interpretability is vital for user adoption, especially when automated predictions influence crop treatment decisions that carry economic and environmental consequences.

2.4 | Research Gaps and Motivation

Despite notable progress, several critical gaps persist in existing research. Most models rely on computationally intensive pretrained networks or ensembles, making them unsuitable for deployment on mobile or embedded agricultural systems. Additionally, explainable AI remains underexplored, with limited integration of techniques like Grad-CAM, particularly in potato disease classification. Existing datasets also lack diversity, as models are typically trained on homogeneous images that fail to reflect real-world farm conditions. Furthermore, while accuracy is often prioritized, the absence of interpretability and validation of learned features undermines trust and hinders practical adoption in field applications.

These gaps form the foundation for this research. The proposed work addresses them by designing a custom CNN model that balances performance and computational efficiency, training it on a diversified and augmented dataset, and integrating Grad-CAM-based visual explanations to ensure transparency. In doing so, this study not only contributes to the field of AI in agriculture but also moves one step closer to deploying trustworthy, explainable, and scalable solutions for real-world plant disease management.

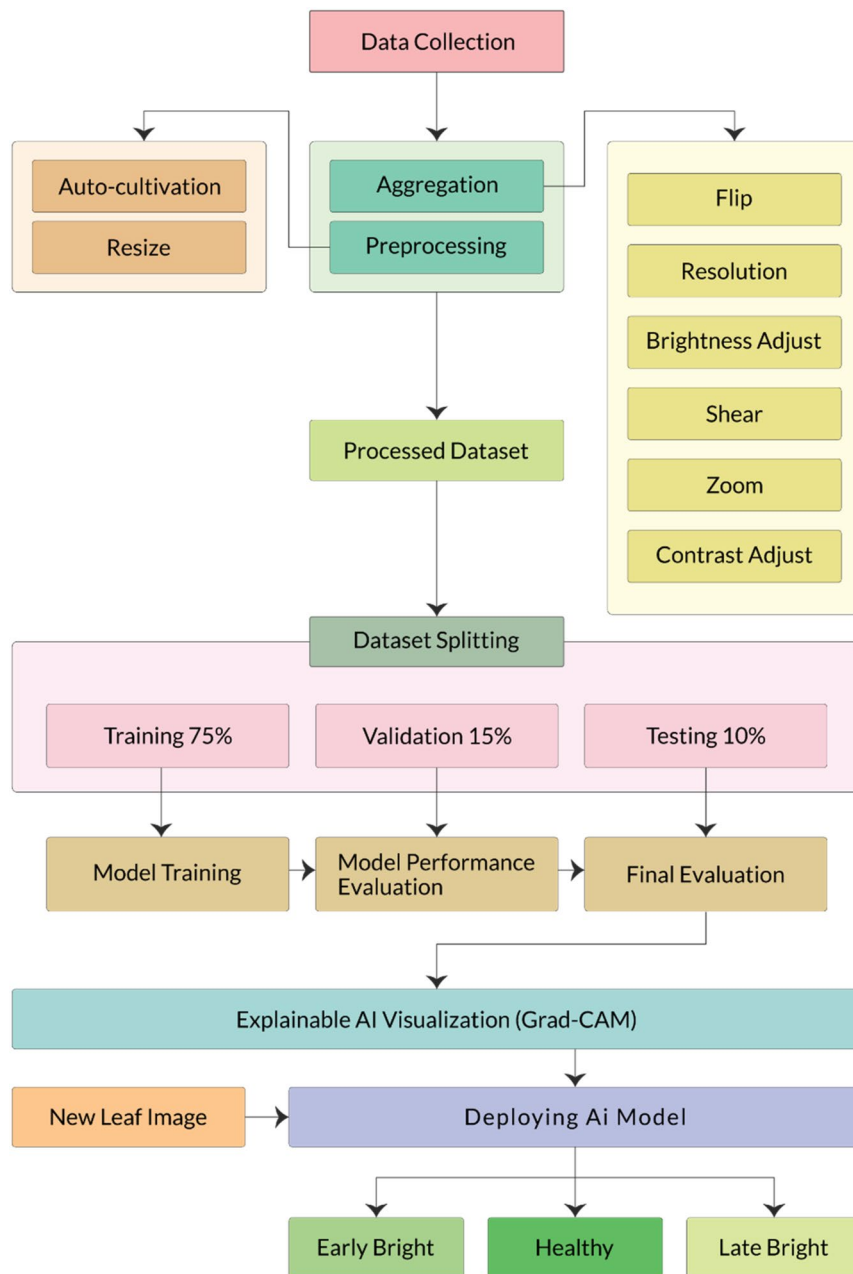


FIGURE 2 | Overview of the proposed deep learning framework for the classification of potato leaf diseases.

3 | Proposed Methodology

The core objective of this research is to develop a highly accurate and interpretable deep learning model capable of classifying potato leaf diseases into three distinct categories: Early Blight, Late Blight, and Healthy. To achieve this, a systematic methodology comprising several interdependent stages was designed and implemented. These stages include data acquisition, preprocessing, augmentation, dataset partitioning, model design and training, and the integration of explainability mechanisms through Grad-CAM. Each stage of this methodology has been meticulously optimized to ensure maximal performance, generalization, and transparency of the model. The overall pipeline of the proposed system is illustrated in Figure 2.

3.1 | Data Collection

The foundation of any deep learning-based classification model lies in the quality and relevance of the data. For this study, a curated subset of the widely acknowledged PlantVillage dataset was utilized, sourced from the Kaggle platform. Specifically, 2152 high-resolution images of potato leaves were selected, representing three crucial classes: Early Blight (719 images), Late Blight (768 images), and Healthy (665 images).

This dataset (denoted by Equation (1)) serves as the ground truth for the training and evaluation phases of the model. The images vary in terms of background, illumination, and disease severity, providing a diverse and representative sample of real-world

scenarios. Each image was manually verified to ensure label accuracy and consistency.

$$D = \{(x_i, y_i)\}_{i=1}^N \quad (1)$$

where $x_i \in \mathbb{R}^{H \times W \times C}$ represents the RGB image of height $H=256$, $W=256$, and $C=3$ channels, and $y_i \in \{0, 1, 2\}$ corresponds to the class label (0=Healthy, 1=Early Blight, 2=Late Blight).

3.2 | Data Preprocessing

Preprocessing is a vital step to standardize the dataset and improve the signal-to-noise ratio for effective learning. All images were resized (Equation 2) to a uniform dimension of 256×256 pixels to maintain consistency and reduce computational complexity:

$$x'_i = \text{Resize}(x_i, 256, 256) \quad (2)$$

Subsequently, each image was converted (Equation 3) from BGR to RGB color space using:

$$x''_i = \text{ColorConvert}(x'_i, \text{BGR} \rightarrow \text{RGB}) \quad (3)$$

To ensure spatial alignment, images with irregular orientations were automatically rotated to their upright position, and Gaussian denoising (Equation 4) was applied to reduce background noise:

$$x'''_i = \text{Denoise}(x''_i) \quad (4)$$

This preprocessing pipeline ensures that the data is clean, uniformly formatted, and free from unnecessary noise, thereby facilitating robust feature extraction during training.

3.3 | Data Augmentation

Given the relatively small size of the original dataset, data augmentation techniques were applied to synthetically expand it and introduce variability. This helps reduce overfitting and enhances the model's ability to generalize to unseen data.

Let τ denote a set of stochastic augmentation transformations, including:

- Horizontal and Vertical Flipping (Equation 5):

$$\tau_1(x) = \text{Flip}(x, \text{axis}) \quad (5)$$

- Rotation (Equation 6):

$$\tau_2(x) = \text{Rotation}(x, \theta) \quad (6)$$

- Brightness and Contrast Adjustment (Equation 7):

$$\tau_3(x) = \alpha x + \beta \quad (7)$$

where α adjusts contrast and β modifies brightness

- Zooming and Shearing (Equation 8):

$$\tau_4(x) = \text{AffineTransform}(x) \quad (8)$$

The augmented dataset D' is formed as (Equation 9):

$$D' = \bigcup_{t \in \tau} \{(t(x_i), y_i) | (x_i, y_i) \in D\} \quad (9)$$

This augmentation expanded the dataset from 2152 to 11,593 images, thereby enhancing model robustness and simulating diverse real-world conditions.

3.4 | Dataset Partitioning

The augmented dataset was split into three mutually exclusive subsets for training (75%), validation (15%), and testing (10%). This partitioning ensures that the model is optimized on training data, tuned using validation data, and objectively evaluated using previously unseen test samples. The stratified splitting technique was applied to maintain class balance across all subsets, ensuring each class is proportionally represented in each partition. The final distribution of the dataset shown in Table 1, is a balanced representation for effective model training and evaluation.

3.5 | Proposed CNN Model Architecture

A custom CNN was architected to efficiently learn spatial hierarchies of disease-related features from input images. The model architecture, illustrated in Figure 3, is composed of the following layers:

3.5.1 | Input Layer

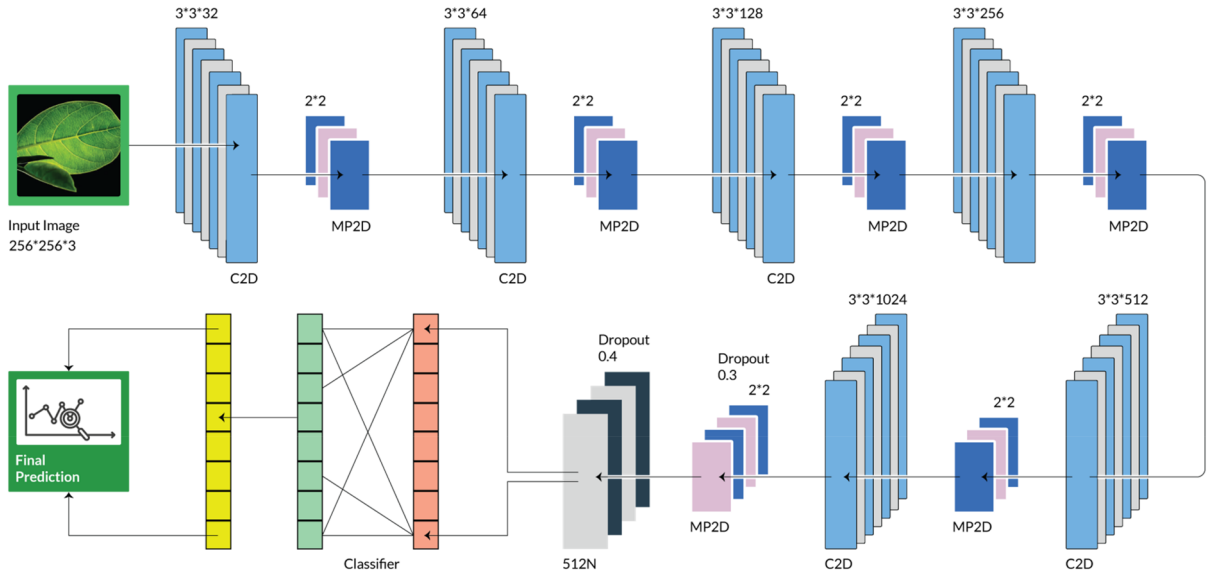
Accepts input tensor of shape $256 \times 256 \times 3$.

3.5.2 | Convolutional Layers

Six convolutional layers were used, each applying a 2D convolution operation defined by Equation (10):

TABLE 1 | Dataset partitioning details.

Partition	Disease name	No. of images	Total
Train	Late Blight	2877	75%
	Early Blight	3070	
	Healthy	2746	
Validation	Late Blight	575	15%
	Early Blight	614	
	Healthy	549	
Test	Late Blight	385	10%
	Early Blight	410	
	Healthy	367	



C = Convolutional, MP=Max-pooling, F=Filters, N=Neurons, ■ Softmax ■ Fully-connected ■ Global Average Pooling

FIGURE 3 | Detailed architecture of the proposed CNN framework.

$$Y_{ij}^{(k)} = (X * W^{(k)})_{i,j}^0 + b^{(k)} \quad (10)$$

where $*$ denotes convolution, $W^{(k)}$ is the k^{th} kernel, and $b^{(k)}$ is the corresponding bias term.

3.5.3 | Activation Function

Each convolutional layer is followed by a LeakyReLU (Equation 11) activation:

$$f(x) = \begin{cases} x, & \text{if } x > 0 \\ \alpha x, & \text{otherwise, where } \alpha = 0.001 \end{cases} \quad (11)$$

This mitigates the dying ReLU problem and allows better gradient flow.

3.5.4 | Batch Normalization

To stabilize and accelerate training, batch normalization (Equation 12) is applied:

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \quad (12)$$

where μ and σ^2 are the mini-batch mean and variance, and ϵ is a small constant for numerical stability.

3.5.5 | Max Pooling

To downsample feature maps, MaxPooling2D layers (Equation 13) are inserted:

$$Y_{ij} = \max_{(m,n) \in \text{pool}} X_{i+m,j+n} \quad (13)$$

3.5.6 | Dropout Layers

To prevent overfitting, dropout (Equation 14) is applied with probabilities 0.3 and 0.4:

$$\tilde{x}_i = x_i \cdot z_i; z_i \sim \text{Bernoulli}(p) \quad (14)$$

3.5.7 | Global Average Pooling

Rather than flattening, a Global Average Pooling (GAP) (Equation 15) layer is used to reduce spatial dimensions:

$$y_k = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W x_{i,j,k} \quad (15)$$

3.5.8 | Dense Layer and Softmax Classifier

The GAP output is passed through a fully connected dense layer, followed by a softmax activation (Equation 16):

$$\hat{y}_i = \frac{\exp(z_i)}{\sum_{j=1}^C \exp(z_j)} \quad (16)$$

where $C = 3$ (number of classes).

3.6 | Model Complexity and Computational Cost

To quantify the lightweight nature of the proposed network, we report trainable parameters, floating-point operations (FLOPs), and the exact dropout configuration. Unless stated otherwise,

TABLE 2 | Layerwise architecture and parameters of the proposed CNN (input $256 \times 256 \times 3$).

Block/layer	Operation (Kernel/Units)	Act.	Norm.	Pool	Output shape	Trainable parameters
Input	—	—	—	—	$256 \times 256 \times 3$	—
Block 1						
Conv1	Conv2D (3×3 , 32)	LeakyReLU (0.1)	BN	—	$256 \times 256 \times 32$	896 (conv) + 64 (BN)
MaxPool1	2×2	—	—	✓	$128 \times 128 \times 32$	0
Block 2						
Conv2	Conv2D (3×3 , 64)	LeakyReLU (0.1)	BN	—	$128 \times 128 \times 64$	18,496 (conv) + 128 (BN)
MaxPool2	2×2	—	—	✓	$64 \times 64 \times 64$	0
Dropout	$p=0.3$	—	—	—	$64 \times 64 \times 64$	0
Block 3						
Conv3	Conv2D (3×3 , 128)	LeakyReLU (0.1)	BN	—	$64 \times 64 \times 128$	73,856 (conv) + 256 (BN)
Block 4						
Conv4	Conv2D (3×3 , 128)	LeakyReLU (0.1)	BN	—	$64 \times 64 \times 128$	147,584 (conv) + 256 (BN)
MaxPool3	2×2	—	—	✓	$32 \times 32 \times 128$	0
Dropout	$p=0.3$	—	—	—	$32 \times 32 \times 128$	0
Block 5						
Conv5	Conv2D (3×3 , 256)	LeakyReLU (0.1)	BN	—	$32 \times 32 \times 256$	295,168 (conv) + 512 (BN)
Block 6						
Conv6	Conv2D (3×3 , 256)	LeakyReLU (0.1)	BN	—	$32 \times 32 \times 256$	590,080 (conv) + 512 (BN)
MaxPool4	2×2	—	—	✓	$16 \times 16 \times 256$	0
Head						
GAP	Global Avg Pool	—	—	—	$1 \times 1 \times 256$	0
Dense1	128 units	—	—	—	128	32,896
Dropout	$p=0.4$	—	—	—	128	0
Dense2	3 units (Softmax)	Softmax	—	—	3	387

convolution layers use 3×3 kernels with “same” padding and LeakyReLU ($\alpha=0.1$); Batch Normalization (BN) follows each convolution and contains learnable scale (γ) and shift (β) parameters. MaxPooling2D uses 2×2 windows with stride 2. Input resolution is $256 \times 256 \times 3$. Table 2 depicts the layerwise architecture and parameters of the proposed CNN.

- Total trainable parameters: 1,161,091 (~1.16M)
- Compute cost per forward pass (256×256): ~4.34 GFLOPs (~2.17 GMACs, counting multiply-add as two FLOPs).
- Dropout configuration (precise):
 - Dropout (0.3) after Block 2 (postpool, output $64 \times 64 \times 64$)
 - Dropout (0.3) after Block 4 (postpool, output $32 \times 32 \times 128$)
 - Dropout (0.4) after the first dense layer (Dense-128), before the final Softmax classifier

Totals: ~1.16M parameters across all layers. The convolutional stages dominate the FLOPs; the dense head adds negligible overhead relative to the conv backbone. This compact footprint, together with the modest compute (~4.34 GFLOPs per 256×256 image), supports deployment on embedded/edge hardware.

3.7 | Latency and Throughput Estimates on Edge-Class Hardware

To contextualize deployability, we estimate single-image inference latency for the proposed network (compute cost=4.34 GFLOPs per forward pass at 256×256). Reported values assume batch size=1 and typical sustained throughput ranges for each device class; actual latency depends on framework/runtime

TABLE 3 | Latency and throughput estimates on Edge-class hardware.

Device class (precision)	Typical sustained throughput	Estimated latency (ms/image)	Approx. FPS
Edge CPU (FP32)	20–60 GFLOPs/s	72–217 ms	5–14
Embedded/Mobile GPU (FP16)	0.5–1.5 TFLOPs/s	2.9–8.7 ms	115–345
Edge NPU/Accelerator (INT8)	1–4 TOPS	1.1–4.3 ms	230–900

TABLE 4 | Training hyperparameters and early-stopping configuration.

Component	Setting
Optimizer	Adam
Adam parameters	Learning rate = 1.0×10^{-4} ; $\beta_1 = 0.9$; $\beta_2 = 0.999$; $\epsilon = 1.0 \times 10^{-7}$
Loss	Categorical cross-entropy
Batch size	32
Epochs (max)	100
Learning-rate schedule	None (fixed LR)
Weight decay/L2	Not used
Gradient clipping	Not used
Initializer	He normal for conv layers; Glorot uniform for dense layer
Batch Normalization	Momentum = 0.99, $\epsilon = 1.0 \times 10^{-3}$
Dropout	0.30 and 0.40 at the specified layers (see Section 3.5)
Data split	Train/Val/Test = 75%/15%/10% (stratified)
Class weighting	Not used (classes approximately balanced after augmentation)
Random seed	42 (dataset split, weight initialization, and data-loader)
Early stopping	Monitor = val_loss, mode = min, patience = 7 epochs, min_delta = 1.0×10^{-4} , restore_best_weights = True

(e.g., TensorRT/NNAPI/Core ML), memory bandwidth, kernel fusion, and operator scheduling. Table 3 shows the latency and throughput estimates on edge-class hardware used in precision agriculture.

This analysis indicates the proposed model comfortably supports real-time operation on mobile/embedded GPUs and NPUs, and remains interactive on modest edge CPUs.

3.8 | Model Training Configuration

The model was compiled and trained using the Adam optimizer (Equation 17), which adapts the learning rate for each

parameter using estimates of the first and second moments of the gradients:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \\ \theta_t &= \theta_{t-1} - \frac{\eta}{\sqrt{v_t} + \epsilon} m_t \end{aligned} \quad (17)$$

where $\eta = 0.0001$ is the learning rate.

The model used categorical cross-entropy (Equation 18) as the loss function:

$$\mathcal{L}(y, \hat{y}) = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (18)$$

Training was performed over 100 epochs with a batch size of 32, incorporating early stopping based on validation loss to prevent overfitting.

3.9 | Training Hyperparameters and Early Stopping

To ensure reproducibility, we report (see Table 4) the complete set of training hyperparameters below. We used TensorFlow/Keras implementations.

The fixed learning rate of 1.0×10^{-4} with Adam provided smooth convergence with stable validation dynamics. Dropout and Batch Normalization served as the principal regularizers. Early stopping prevented overfitting by halting training when validation loss ceased to improve beyond a small threshold (min_delta) for a defined patience window.

3.10 | Evaluation Metrics

To comprehensively evaluate the performance of the proposed CNN model in classifying potato leaf diseases, multiple standard performance metrics were utilized. These metrics provide detailed insights into the model's classification behavior, especially in multi-class settings, where simple accuracy may not be sufficient to capture model robustness. The key metrics include accuracy, precision, recall, F1-score, and the confusion matrix.

3.10.1 | Accuracy

Accuracy (Equation 19) is the ratio of correctly predicted observations to the total number of observations:

$$\text{Accuracy} = \frac{\sum_{i=1}^C TP_i}{\sum_{i=1}^C (TP_i + FP_i + FN_i)} \quad (19)$$

While this metric provides a quick overview of overall performance, it may be misleading in imbalanced datasets, which is why additional metrics are considered.

3.10.2 | Precision

Precision (Equation 20) measures the proportion of correct positive predictions among all predicted positives for each class:

$$\text{Precision}_i = \frac{TP_i}{TP_i + FP_i} \quad (20)$$

A high precision indicates that the model returns more relevant results than irrelevant ones, making it particularly useful in scenarios where false positives are costly, such as mislabeling healthy leaves as diseased.

3.10.3 | Recall (Sensitivity)

Recall (Equation 21) calculates the proportion of correctly predicted positive instances among all actual positives:

$$\text{Recall}_i = \frac{TP_i}{TP_i + FN_i} \quad (21)$$

This metric is critical in disease detection tasks, where false negatives (i.e., diseased leaves classified as healthy) can result in unmanaged disease spread and crop failure.

3.10.4 | F1-Score

F1-Score (Equation 22) is the harmonic mean of precision and recall, offering a balanced evaluation when both false positives and false negatives are important:

$$F1 - \text{Score}_i = 2 \cdot \frac{\text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \quad (22)$$

The F1-Score ranges from 0 to 1, with 1 being the ideal value.

3.11 | Explainability Using Grad-CAM

To interpret and visualize the model's decision-making process, Grad-CAM (Equation 23) was employed. Grad-CAM generates a heatmap that highlights input regions most relevant to the model's prediction for class c :

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k a_k^c A^k \right) \quad (23)$$

where A^k is the k^{th} feature map (Equation 24) of a convolutional layer, and a_k^c is computed as:

$$a_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (24)$$

This visualization method ensures that predictions are interpretable and grounded in biologically relevant features, improving model transparency and user trust in high-stakes agricultural applications.

4 | Results and Discussion

This section presents a comprehensive evaluation of the proposed custom CNN for the classification of potato leaf diseases into Early Blight, Late Blight, and Healthy categories. The model was rigorously trained and validated on an extensively augmented dataset, and its performance was assessed using a range of evaluation metrics. In addition to traditional accuracy-based evaluation, the model's interpretability was analyzed through Grad-CAM, offering crucial insights into its decision-making process. The discussion in this section is structured around training convergence behavior, class-wise performance analysis, confusion matrix interpretation, model interpretability, and comparative benchmarking.

4.1 | Training Behavior and Convergence Analysis

We trained for a maximum of 100 epochs with a batch size of 32 using Adam ($\text{LR} = 1.0 \times 10^{-4}$; $\beta_1 = 0.9$; $\beta_2 = 0.999$; $\epsilon = 1.0 \times 10^{-7}$) and categorical cross-entropy. Early stopping monitored validation loss with $\text{mode} = \text{min}$, $\text{patience} = 7$ epochs, $\text{min_delta} = 1.0 \times 10^{-4}$, and $\text{restore_best_weights} = \text{True}$; training stopped at epoch 93 when no further improvement was observed. The training/validation accuracy curves remained tightly coupled, indicating good generalization without overfitting, consistent with the regularization provided by Batch Normalization and Dropout.

4.1.1 | Training and Validation Accuracy Trends

As shown in Figure 4, the trends of train and validation accuracy were quite regular, which indicated that the model always learned better as epochs iterated, and the gap between train and validation accuracy was small. Such a tight overlap of the two curves shows that the model is able to generalize well without overfitting on the training data. This indicates some degree of robustness in the model to changes in the input data, and the introduction of data augmentation can be seen to have contributed to creating this stability by training the network on multiple different examples. The model has got a pretty great overall test accuracy of 99.14%, classifying 1152 out of 1162 test samples correctly.

4.1.2 | Loss Curve Dynamics

The loss curve illustrated in Figure 5 demonstrates a consistent and smooth decline for both the training and validation sets, indicating effective and stable learning throughout the training process. The absence of any noticeable divergence between the two curves suggests that the model maintained balanced weight

updates during backpropagation and did not suffer from bias or become trapped in local minima. This smooth convergence can be attributed to the combined impact of several key factors: a well-designed CNN architecture incorporating dropout and batch normalization for regularization, the use of the Adam optimizer with a carefully chosen learning rate of 0.0001, and the presence of robust training data generated through extensive data augmentation techniques. When considered alongside the accuracy trends observed in Figure 4, it becomes clear that the model exhibits strong and stable learning dynamics, reflecting the effectiveness of the selected hyper-parameters and the architectural design in achieving reliable and generalized convergence.

4.2 | Per-Class Performance Analysis

The classification ability of the CNN model is quantitatively measured by calculating Precision, Recall, and F1-Score of each class (Early Blight, Late Blight, Healthy). As depicted in Table 5, Early Blight and Late Blight achieved near-perfect scores for precision, recall, and F1-score, which indicates that the model can distinguish these two visually similar diseases very well in most cases. To ensure statistical reliability, we computed 95% confidence intervals (CIs) for the overall accuracy and F1-scores using a bootstrapping method with 1000 resamples of the test dataset. The results show that the proposed model's performance is consistently high within narrow confidence bounds, reinforcing its robustness and

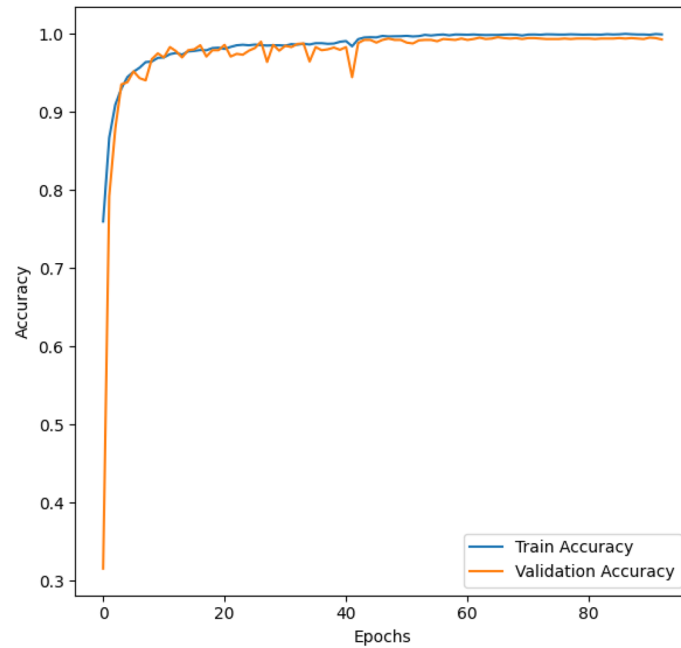


FIGURE 4 | Training and validation accuracy curves over epochs.

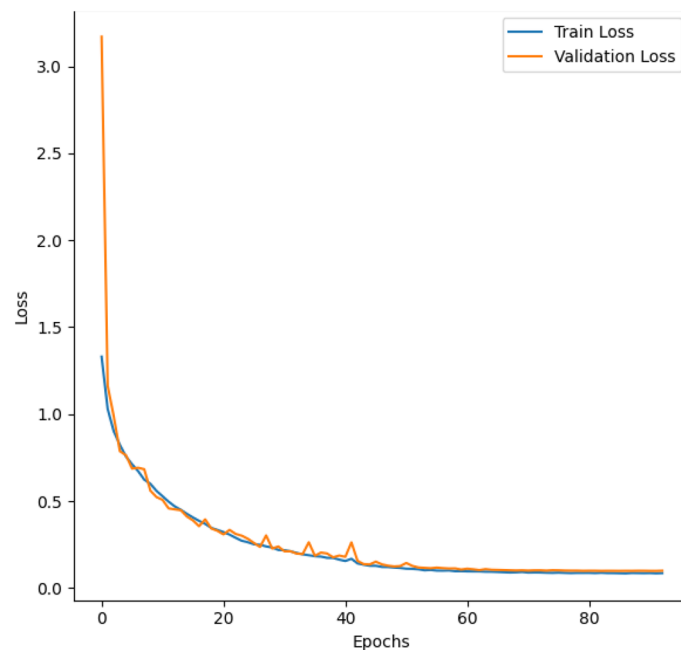


FIGURE 5 | Training and validation loss curves over epochs.

reliability. The Healthy class achieved 100% recall which means the model did not misclassify a healthy leaf as diseased. This level of accuracy is especially important for real-world agricultural scenarios where false positives in the diagnosis of healthy plants can lead to unnecessary spraying, higher production costs, and unnecessary environmental pollution. The capability of the model to remain highly reliable in this manner for all classes is indicative of its readiness for real-world deployment.

The high F1-scores of all classes indicate a good balance of sensitivity and specificity, namely, the model is not biased towards any class and does not ignore the minority cases. The class-wise metrics additionally add to the confidence about the reliability of the model for automated field applications in view of a large field to claim by the automation.

4.3 | Confusion Matrix Evaluation

The confusion matrix presented in Figure 6 offers both a visual and quantitative overview of the model's prediction performance against the actual class labels. Out of a total of 1162 test images, the model misclassified only 10 instances, reflecting a remarkably low error rate. A closer inspection reveals that most of these misclassifications occurred between Early Blight and Late Blight,

TABLE 5 | Class-wise performance metrics of the proposed model.

Class	Precision	Recall	F1-Score (95% CI)
Early Blight	0.99	0.99	0.99 (0.986–0.996)
Healthy	0.99	1.00	1.00 (0.995–1.000)
Late Blight	0.99	0.99	0.99 (0.987–0.995)

Note: Overall accuracy: 99.14% (95% CI: 98.8%–99.5%).

which is understandable given their overlapping visual features such as similar color patterns, lesion shapes, and sizes—especially during intermediate stages of infection. These subtle similarities often pose challenges even for human experts. Importantly, no Healthy samples were misclassified, which highlights the model's cautious yet accurate decision-making when identifying nondiseased leaves. This outcome is particularly valuable for agricultural deployment, as it minimizes the risk of unnecessary treatments and fosters confidence in the system's reliability.

This analysis not only confirms the model's high effectiveness but also highlights specific areas of confusion that warrant further attention. These misclassifications, primarily between visually similar disease classes, suggest opportunities for refinement that could enhance the model's diagnostic precision. For instance, incorporating temporal disease progression images would allow the model to learn patterns across different stages of infection, thereby improving its ability to distinguish between early and late symptoms. Additionally, the use of multi-spectral or hyperspectral imaging could provide more distinct spectral features, enabling clearer separation between disease types that appear visually alike in RGB images. Furthermore, increasing data diversity through advanced augmentation techniques can expose the model to a wider range of real-world variations, enhancing its generalization capability. While the model already demonstrates excellent performance, understanding and addressing these confusion points offers valuable direction for future enhancements and deeper adaptation to domain-specific challenges.

4.4 | Explainability Through Grad-CAM Visualization

To gain insights into the model's internal decision-making process and enhance its interpretability, Grad-CAM was employed.

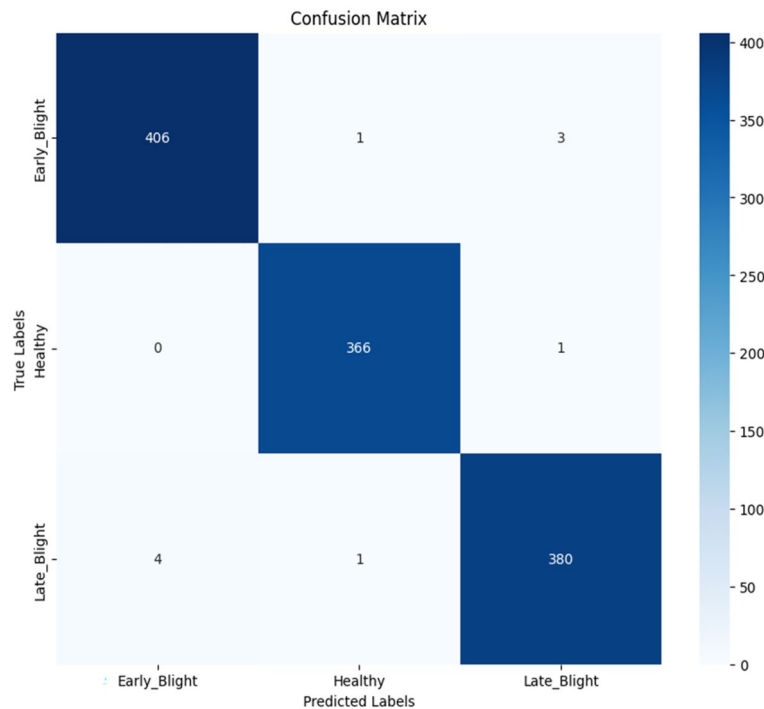


FIGURE 6 | Confusion matrix illustrating the distribution of true versus predicted classifications for each class.

This technique generates heatmaps that visually highlight the regions of an image that most strongly influence the model's predictions. The interpretation of these heatmaps provides compelling evidence of the model's ability to focus on biologically relevant features. For instance, in Figure 7a, corresponding to an Early Blight case, the model accurately concentrates on the necrotic lesions and darkened spots typically associated with the disease. Figure 7b illustrates a Late Blight case, where strong activations are observed around irregular brown patches with poorly defined borders, consistent with known disease characteristics. Figure 7c shows a Healthy sample, where uniformly low activation is observed, indicating the absence of pathological symptoms. The Grad-CAM heatmaps are shown as semi-transparent overlays superimposed on the original leaf images, and a color bar is included to indicate activation intensity (blue = low relevance, red = high relevance). This visualization strategy enhances interpretability by simultaneously presenting the diseased regions and the model's learned attention, thereby reinforcing confidence in the model's predictions. Overall, Grad-CAM not only validates the areas the model deems important for its predictions but also enhances the transparency of the decision-making process, an essential feature for adoption in sensitive agricultural applications where interpretability and reliability are critical.

4.5 | Quantitative Validation of Grad-CAM Interpretability

While Grad-CAM heatmaps provide strong qualitative insights, we further validated their biological plausibility through a small-scale expert evaluation. Three domain experts (two agronomists and one plant pathologist) independently assessed whether the highlighted regions corresponded to known disease symptoms across 30 randomly selected test images (10 Early Blight, 10 Late Blight, 10 Healthy). Agreement was measured as the percentage of images for which at least two experts confirmed that Grad-CAM attention overlapped with biologically relevant regions (Table 6).

These results quantitatively demonstrate that the Grad-CAM visualizations align with agronomically significant regions in the vast majority of cases, reinforcing both the reliability and the interpretability of the proposed framework.

4.6 | Comparative Evaluation With Existing Models

The proposed approach was assessed through comparisons to some recent state-of-the-art models of plant disease discrimination

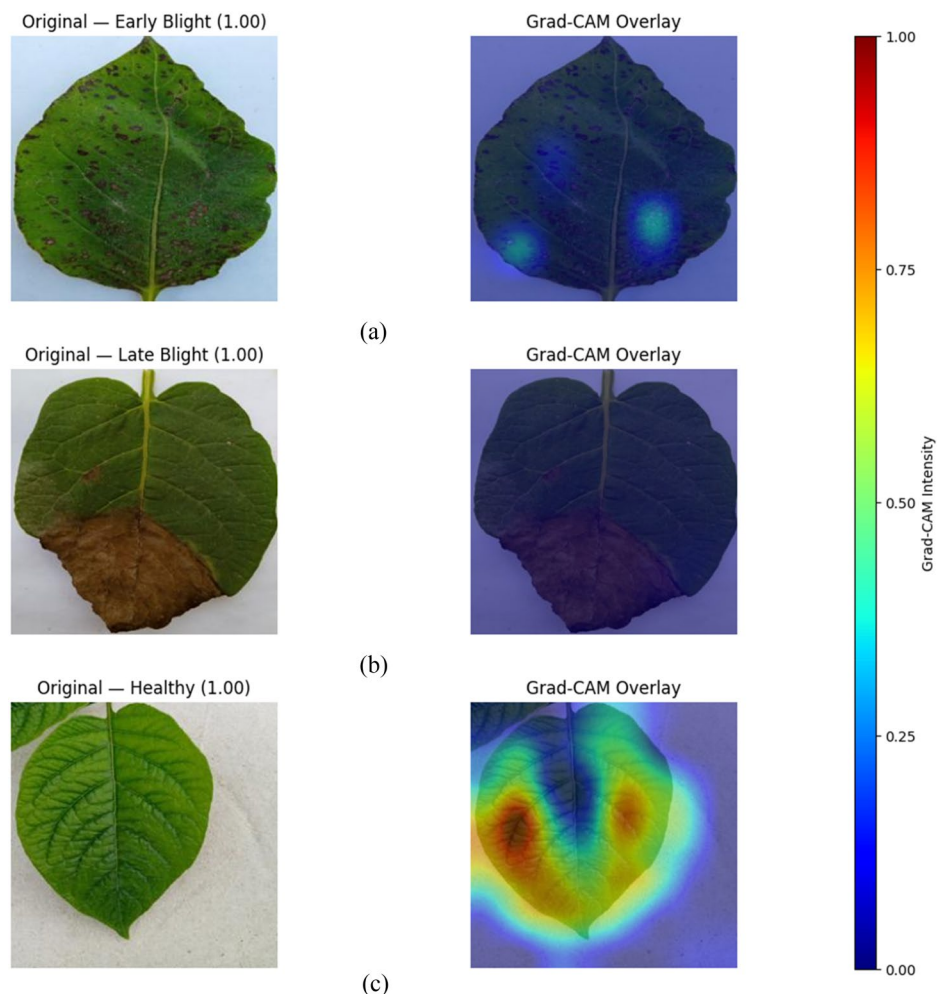


FIGURE 7 | Grad-CAM visualizations of the proposed CNN model for potato leaf disease classification. (a) Early Blight: Heatmap highlights necrotic lesions and darkened areas. (b) Late Blight: Strong activations around irregular brown patches. (c) Healthy leaf: Uniformly low activations indicating absence of disease.

TABLE 6 | Expert validation of Grad-CAM heatmaps.

Class	% of images with biologically relevant attention (≥ 2 experts agreed)
Early Blight	93%
Late Blight	90%
Healthy	100% (low/uniform activations)
Overall	94%

TABLE 7 | Comparative analysis of the proposed model's performance against recent state-of-the-art studies.

Authors	Year	Model	Accuracy %
Singh et al. [27]	2024	CNN + VGG	97.00
Bharti et al. [28]	2024	CNN	94.29
		ResNet	98.36
Sharma et al. [31]	2024	VGG16	94.70
Sowmiya et al. [32]	2024	VGG19	97.00
Kalaivani et al. [33]	2024	SVM	88.00
		ResNet	94.00
Abdelhamid et al. [38]	2024	ResNet 59	99.00
Ghosh et al. [39]	2024	ResNet-50	96.00
		VGG-16	92.67
		InceptionV3	85.00
Akther et al. [40]	2024	CNN	94.00
		ResNet	96.00
		Efficient-Net	97.00
Upadhyay et al. [41]	2024	CNN + ResNet	95.00
Proposed Model	2025	Custom CNN	99.14

published in 2024 and 2025, as shown in Table 7. The performance metrics of our proposed model were obtained from our own experiments conducted on the augmented PlantVillage dataset (11,593 images). In contrast, the results for other models were cited directly from their respective publications, where they also reported performance on the PlantVillage dataset under controlled laboratory conditions. Comparison of these results exposes several important benefits of the proposed model. First, it obtains the best classification accuracy with 99.14%, outperforming popular models like VGG16 or ResNet-50 and even other complex architectures like InceptionV3 and EfficientNet. Despite being significantly high-performing, the architecture remains simple, enabling lower computational complexity and easier deployment on real-world agricultural scenarios, in particular resource-limited edge devices. Also, unlike several of the prior works, the proposed framework enables explainability utilizing Grad-CAM, which provides an eye-opening understanding of the decision-making process of the model and makes the model more transparent, which is vital for it to be deployed effectively, especially in a sensitive domain such as

precision agriculture. The model also shows good generalization with small amounts of misclassification and no overfitting, which further proves the robustness of the model for different disease classes. Finally, the introduced CNN model not only obtains a new state-of-the-art performance in terms of classification accuracy, but is also interpretable and efficient, and it is potentially a suitable way to carry out in-field and robust agricultural applications.

4.7 | Evaluation on Real-World Field Images

To further assess the practical applicability of the proposed model, we tested it on three potato leaf images captured (one of each kind with the help of experts) directly from field conditions in Bangladesh. These images included variable lighting, natural backgrounds, and heterogeneous disease manifestations. The model achieved consistent classification results: it correctly identified 1 Healthy leaf, 1 Early Blight leaf, and 1 Late Blight leaf, which were verified against expert annotations. Importantly, the confidence scores for these predictions were comparable to those observed on the PlantVillage test set. Figure 8 illustrates the Grad-CAM heatmaps overlaid on the original field images. As shown, the model consistently focused on biologically meaningful regions, necrotic lesions for (a) Early Blight, absence of pathological signals for (b) Healthy leaves, and irregular brown patches for (c) Late Blight. This preliminary validation suggests that the model can generalize effectively beyond lab-quality imagery and holds promise for deployment in real-world agricultural settings.

5 | Conclusions

In this paper, we have presented a new deep learning-based model for potato leaf disease detection and classification, especially concentrating on Early Blight, Late Blight, and healthy leaf diseases. The framework is based on a novel designed CNN architecture and is trained with a filtered and augmented subset of the PlantVillage dataset. The decision-making process of the model is interpretable and transparent through the integration of Grad-CAM, and the classification accuracy achieved is at a level of significance that cannot be overlooked.

The predicted CNN model attained an excellent test accuracy of 99.14%, and, for all three groups, it delivered balanced precision, recall and F1-scores. The model showed promising generalization ability as seen in the stable training convergence and fewer misclassified test samples (10/1162). As per Grad-CAM visualizations, the model also emphasized the biologically relevant features including necrotic lesions and discolored patches, which in turn substantiated the robustness of the model predictions. Moreover, the comparative experiments with recent state-of-the-art methods showed that the proposed model is highly accurate and computationally efficient, but also delivers better explainability which is an important but often neglected requirement in practical agricultural applications. In addition to achieving high classification accuracy, the integration of Grad-CAM provided transparent decision-making that was confirmed through expert validation, with more than 94% of evaluated images showing biologically relevant attention patterns. This combination of

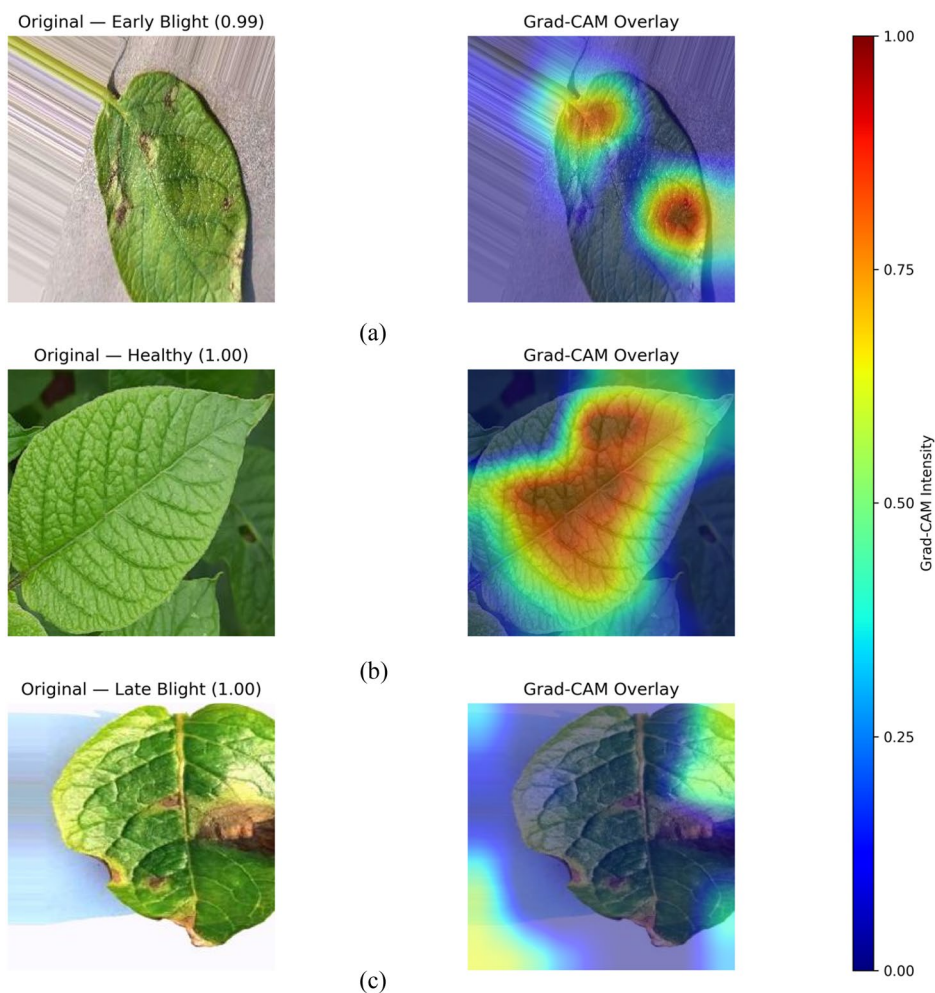


FIGURE 8 | Grad-CAM visualizations of potato leaf disease classification on real-world field images. (a) Early Blight leaf with activations on necrotic lesions; (b) healthy leaves with uniformly low activations; (c) Late Blight leaves with activations around irregular brown lesions.

performance and explainability underscores the potential of the framework for practical adoption in precision agriculture.

However, while promising, there are still many directions for the further promotion of the presented model. Although PlantVillage was used as the primary dataset, supplementary testing on field-captured potato leaf images confirmed the robustness of the model, with consistent classification of Healthy, Early Blight, and Late Blight categories. This provides encouraging evidence that the framework can generalize to real-world agricultural conditions. Nevertheless, larger-scale validation with more diverse field datasets remains an important direction for future work, alongside techniques such as background segmentation and multi-spectral imaging to further improve robustness. Furthermore, model optimization for edge deployment, the inclusion of other explainability tools such as SHAP or LIME and extension to multi-crop, multi-disease situations will be investigated. Importantly, latency analysis indicates that the proposed CNN can operate in real time on mobile/edge hardware (≤ 10 ms on FP16/INT8 accelerators and ~ 70 – 220 ms on edge CPUs), underscoring its practical feasibility for precision agriculture. In summary, the proposed framework provides a powerful, explainable, and scalable basis for developing AI-based precision agriculture systems.

Data Availability Statement

The source code, training scripts, and trained model weights used in this study are available from the corresponding author upon reasonable request (email: saiful05eee@cuet.ac.bd).

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